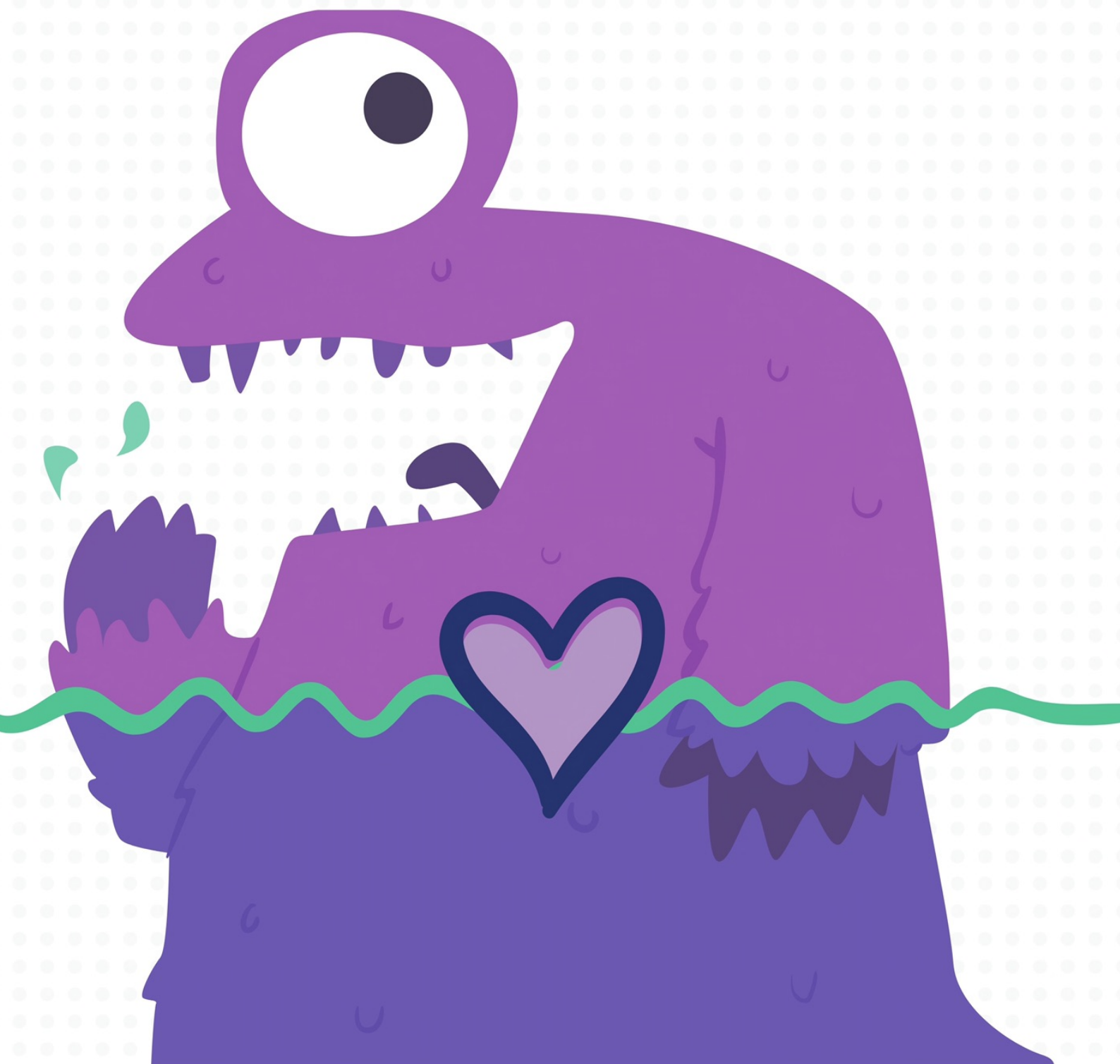


UNDERSTANDING

MURMURS

NOTESPAEDIA REVISION NOTES



Murmurs

MITRAL STENOSIS

- STENOSIS OF MITRAL VALVE

• SEVERE M.S

- MITRAL VALVE AREA $< 1.5 \text{ cm}^2$

• CRITICAL M.S

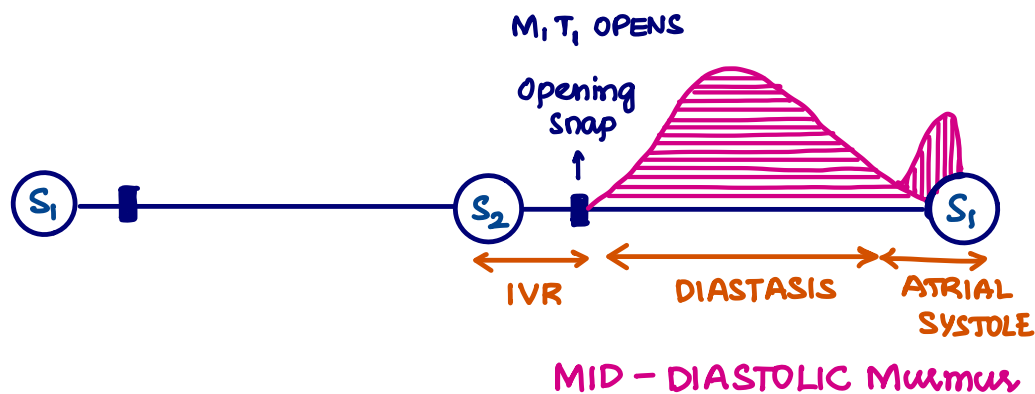
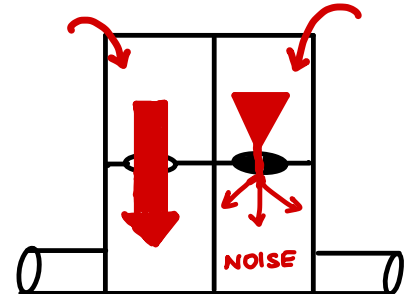
- MITRAL VALVE AREA $< 1 \text{ cm}^2$

• BLOOD comes through STENOSED (NARROW) Valve → CREATES TURBULENCE.

• Mitral valve opens with more force - LOUD OPENING SNAP.

• When Mitral valve opens Blood passes From Left Atria → Left Ventricle

• Hence, Murmur begins after opening of Mitral valve.



• MID DIASTOLIC Murmur is present because murmur starting from OPENING SNAP.

- PEAK IN THE MIDDLE OF DIASTOLE

- SECONDARY PEAK Before S₁ → PRE SYSTOLIC ACCENTUATION is due to ATRIAL SYSTOLE

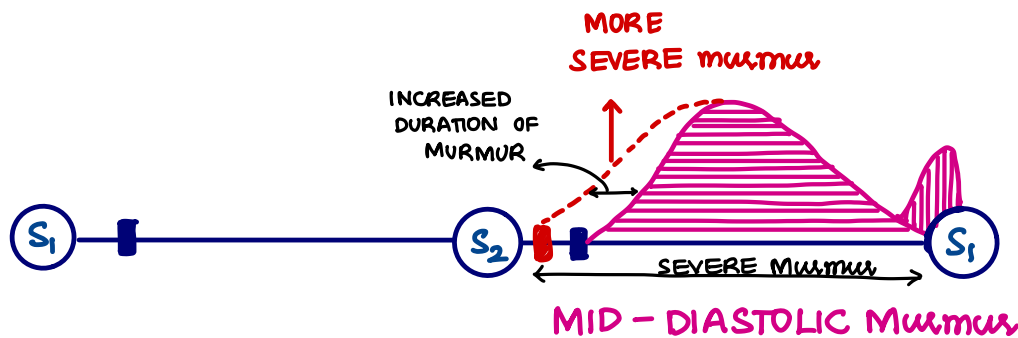
- PRE SYSTOLIC ACCENTUATION DISAPPEARS When ATRIAL SYSTOLE IS NOT HAPPENING.

Eg:- in ATRIAL FIBRILLATION. (AS ATRIA are TWITCHING).

Murmur is BEST HEARD in
LEFT LATERAL DECUBITUS POSITION
(Heart comes close to chest wall)

$$\text{SEVERITY OF M.S} \propto \frac{1}{S_2 - \text{Opening snap gap.}}$$

- SEVERITY OF MITRAL STENOSIS IS DECIDED BY DURATION OF MURMUR (length)
Not By -
- LOUDNESS OF S_1
- LOUDNESS OF P_2
- INTENSITY OF MURMUR.



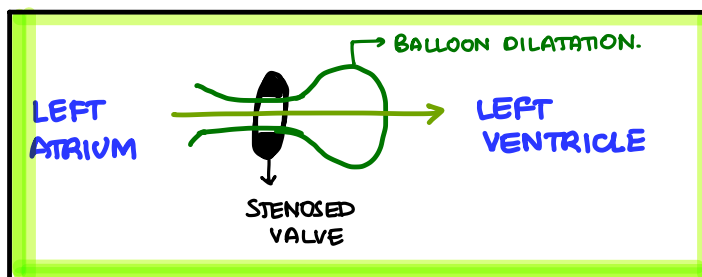
AS SEVERITY OF MITRAL STENOSIS INCREASES, OPENING SNAP OCCURS EARLY AND DURATION OF MURMUR INCREASES.

- SEVERE MS -

Rx: PERCUTANEOUS MITRAL BALLOON VALVOTOMY (PMBV) - T.O.C.

(puncture BASILIC Vein → SUBCLAVIAN Vein → Right ATRIUM

INTER ATRIAL SEPTUM ↓
Left ATRIUM → Left VENTRICLE → STENOSSED VALVE



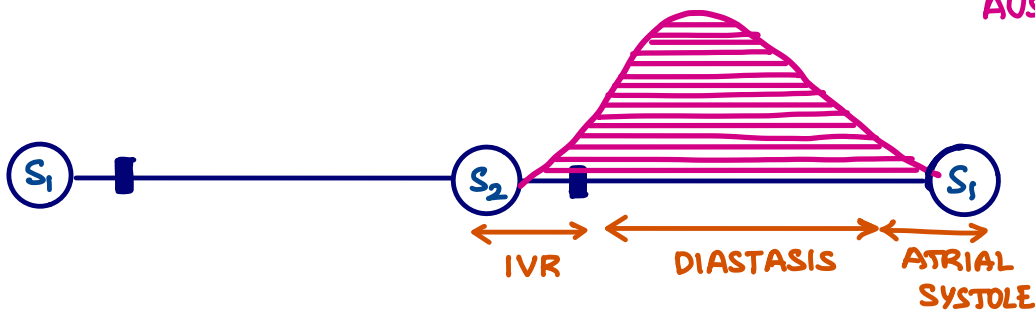
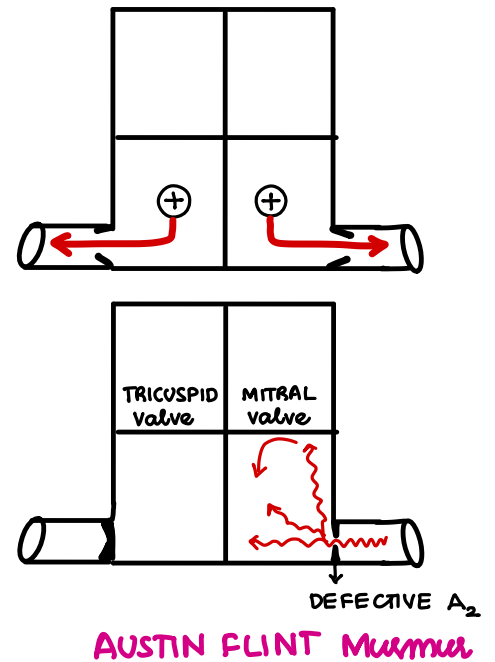
- PMBV IS CONTRAINDICATED in.

- CALCIFIED SEVERE M.S → Valve Breaks into Fragments, Fragments may enter Brain → RISK OF EMBOLIC STROKE
- SEVERE MS + MR → AS MS improves but MR WORSENS.
- SEVERE MS + Left ATRIAL APPENDAGE THROMBI → Emboli may enter SYSTEMIC CIRCULATION.

AORTIC and PULMONIC REGURGITATION :

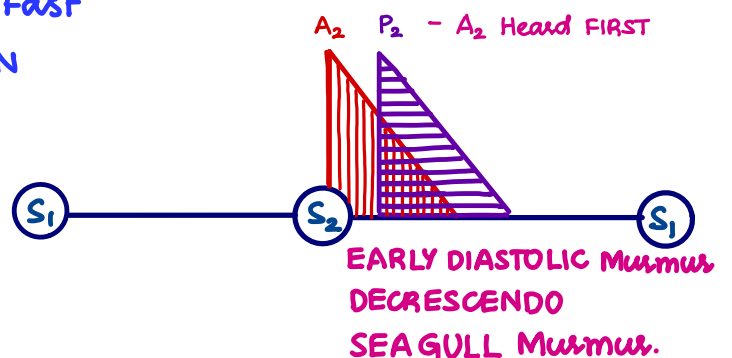
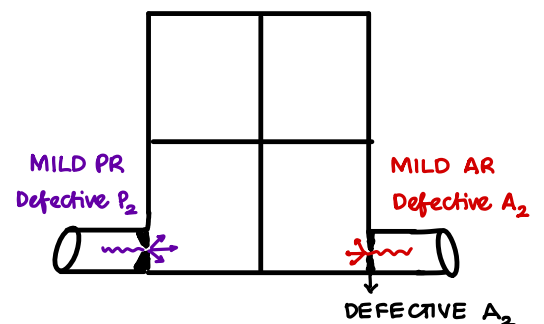
SEVERE AORTIC REGURGITATION

- Blood Leaks back into Left Ventricle and Hits Mitral valve leaflets producing a MID-DIASTOLIC MURMUR
- Also known as AUSTIN FLINT MURMUR
- MURMUR starts FROM S_2 , NOT OPENING SNAP (NOT like M.S)



MILD AORTIC REGURGITATION

- As only Mild, Less Blood leaks into R.V
- Produces LESS NOISE
- Hence, SHORTER DURATION OF MURMUR.
- Same in PULMONIC REGURGITATION.
- Turbulence Begins at $A_2 P_2$ and Dies Fast producing TRIANGULAR CONFIGURATION OF MURMUR.
- EARLY DIASTOLIC MURMUR
As $A_2 P_2$ Doesn't close together and Time lag is 30 msec.
- GRAHAM STEELE MURMUR
in PR in patients with PULMONARY ARTERY HYPERTENSION.



DIASTOLIC MURMUR

EARLY DIASTOLIC Murmur

- G - GRAHAM STEELE
- A - A.R (mild)
- P - P.R (mild)

MID DIASTOLIC Murmur

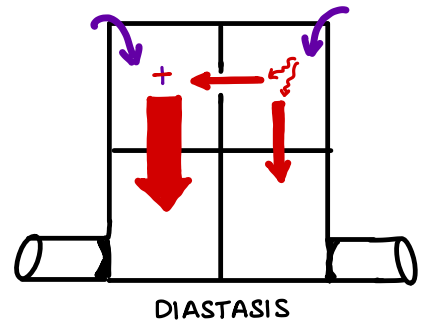
- MITRAL STENOSIS
- AUSTIN FLINT Murmur
- FLOW Murmur

LATE DIASTOLIC Murmur

- CAREY COOMBS Murmur in RHEUMATIC CARDITIS.

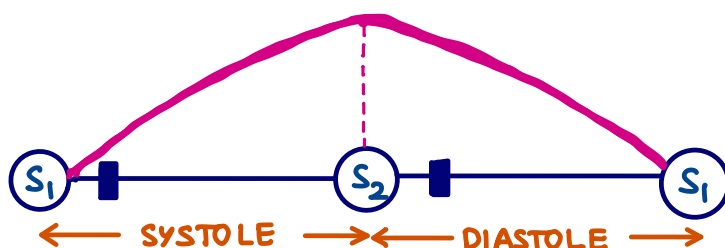
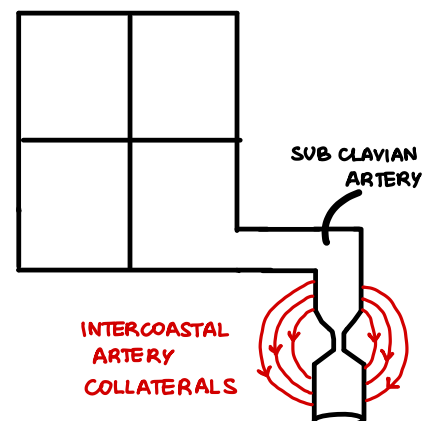
ATRIAL SEPTIC DEFECT

- The Gradient b/w LA and RA (4mmHg) leads to SHUNTING OF BLOOD.
- BUT NO SHUNT Murmur (very less pressure Gradient).
- INCREASED BLOOD FLOW via TRICUSPID valve (Venous Return + Blood via ASD)
- FLOW MURMUR → MID DIASTOLIC MURMUR.
- This Blood Flow occurs during DIASTOLE at DIASTASIS. Hence, causes MID DIASTOLIC MURMUR



COARCTION OF AORTA

- MOST COMMON SITE - DISTAL TO ORIGIN OF SUBCLAVIAN ARTERY (Aortic Isthmus)
- Murmur heard in both SYSTOLE and DIASTOLE (collaterals Formed Create turbulence)
- PRODUCTION of a CONTINUOUS MURMUR.
- PEAK at S₂.



CONTINUOUS MURMUR

CAUSES OF CONTINUOUS MURMUR

1. CoA (COARCTION OF AORTA) :

- TURNER SYNDROME

2. PATENT DUCTUS ARTERIOSUS

- CONGENITAL RUBELLA SYNDROME

3. PREGNANCY

- MAMMARY SOUFFLE

(Due to increased flow of blood via INTERNAL MAMMARY ARTERY)

4. VENOUS HUM

- Due to DILATED VEINS.

5. Rupture of SINUS OF VALSALVA

6. PERIPHERAL PULMONIC STENOSIS

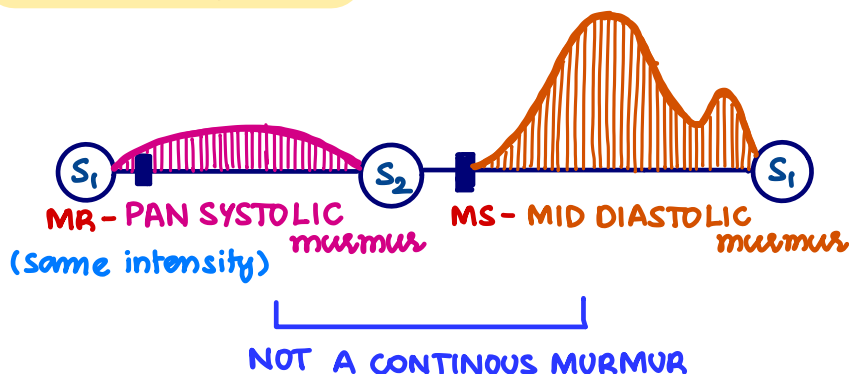
(Not Valvular P.S)

- Narrowing in PA and its Branches
- produces CONTINUOUS MURMUR with Blood Flow.

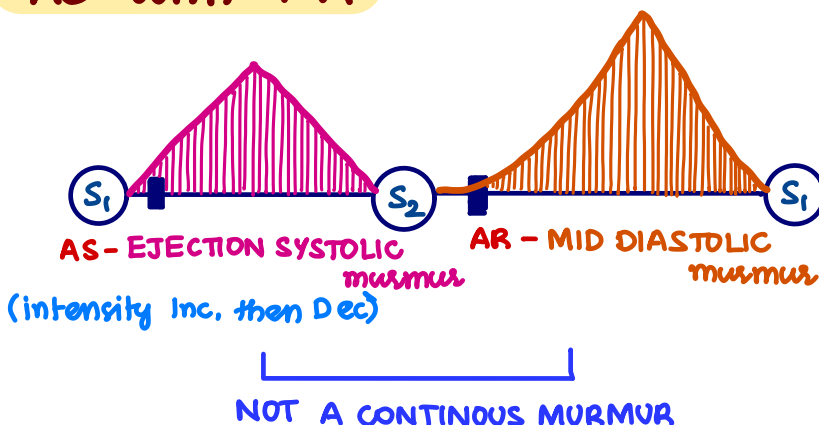
MOST COMMON CARDIAC LESION in TURNER Syn.

1. Bicuspid Aortic valve
2. Coarction of Aorta

MS with MR



AS with AR



CONTINUOUS MURMUR

- in Both Systole & Diastole
- PEAKS AT S₂
- Not Applicable to COMBO LESIONS
 - MS + MR / TS + TR
 - AS + AR / PS + PR

SYSTOLIC MURMURS

SYSTOLIC MURMURS

EJECTION SYSTOLIC Murmur



PEAK at MIDDLE OF SYSTOLE

- Valvular AORTIC STENOSIS
- PULMONIC STENOSIS
- HOCM - Subvalvular Aortic Stenosis.
- TETRALOGY OF FALLOT
 - Subvalvular pulmonic Stenosis

PAN SYSTOLIC Murmur



SAME INTENSITY

- MITRAL REGURGITATION
- TRICUSPID REGURGITATION

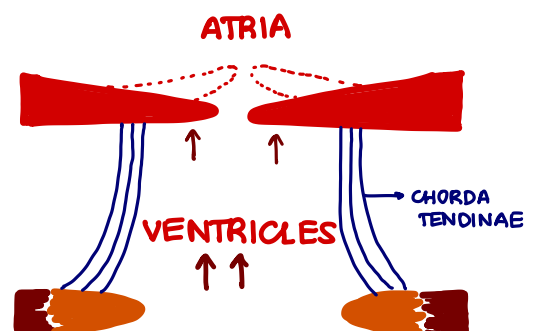
LATE SYSTOLIC Murmur



- MITRAL VALVE PROLAPSE

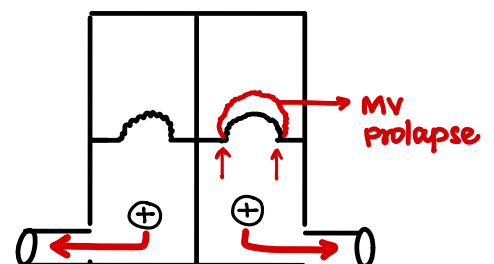
MV PROLAPSE

- Damage to PAPILLARY Muscle
 - ↓
- CHORDA TENDINAE SUPPORT IS LOST
 - ↓
- INCREASED PRESSURE
 - ↓
- VALVE LEAFLETS GO HIGHER THAN Normal



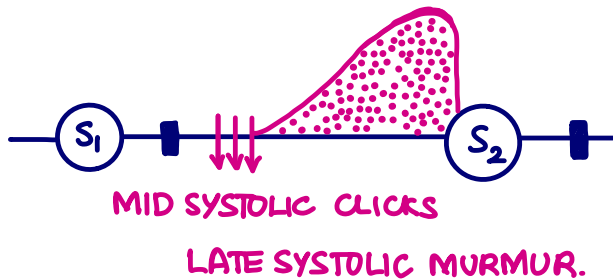
CAUSES

01. Myxomatous Degeneration of papillary Muscles
 - MOST COMMON
02. Aschoff Nodules in Rheumatic Fever
03. Myocardial Infarction (ISCHEMIA)
04. Sub Acute Bacterial Endocarditis (INFECTION)
05. Marfan Syndrome (FIBRILLIN - 1 Defect)



- FEATURES

- EXTRA TENSION in VALVE LEAFLETS / CHORDA TENDINAE
- AUSCULTATORY → MID SYSTOLIC CLICKS (many) - Due to STRETCH in CHORDA TENDINAE.
- COMPLICATION - MITRAL VALVE REGURGITATION.



SUMMARY

• DIASTOLIC MURMUR

- EARLY: GRAHAM STEELE MURMUR (PAH)
- MID : AUSTIN FLINT MURMUR (SEVERE AR)
- LATE : CAREY COOMBS MURMUR (RHEUMATIC CARDITIS)

• CONTINUOUS MURMUR

- COARCTION OF AORTA
- PATENT DUCTUS ARTERIOSUS
- PERIPHERAL PULMONARY STENOSIS.

• SYSTOLIC MURMUR

- EJECTION SYSTOLIC MURMUR - AS / PS / HOCM
- PAN SYSTOLIC MURMUR - MR / TR / USD
- LATE SYSTOLIC MURMUR - MV prolapse.

- ALL MURMURS DECREASE in INTENSITY with VALSALVA / STANDING or AMYL NITRATE INHALATION (Due to DEC. in Venous Return)
 - EXCEPT : HOCM - LOUDER
 - MVP - LONGER

- ALL MURMURS INCREASE in INTENSITY with SITUPS / HAND DRIP or (Due to INC. in PERIPHERAL RESISTANCE)
 - EXCEPT : HOCM - SOFTER
 - MVP - SHORTER